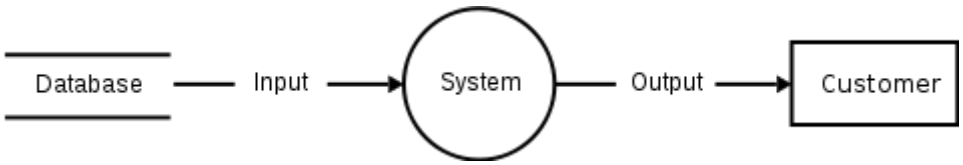
DATA-FLOW DIAGRAM



Data flow diagram with data storage, data flows, function and interface

A **data-flow diagram** (DFD) is a way of representing a flow of a data of a process or a system (usually an information system). The DFD also provides information about the outputs and inputs of each entity and the process itself. A data-flow diagram has no control flow, there are no decision rules and no loops. Specific operations based on the data can be represented by a flowchart.

There are several notations for displaying data-flow diagrams. The notation presented above was described in 1979 by Tom DeMarco as part of Structured Analysis.

For each data flow, at least one of the endpoints (source and / or destination) must exist in a process. The refined representation of a process can be done in another data-flow diagram, which subdivides this process into sub-processes.

The data-flow diagram is part of the structured-analysis modelling tools. When using UML, the activity diagram typically takes over the role of the data-flow diagram. A special form of data-flow plan is a site-oriented data-flow plan.

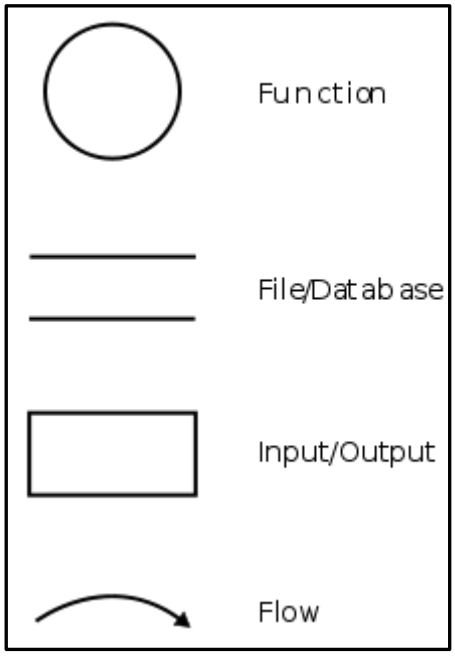
Data-flow diagrams can be regarded as inverted Petri nets, because places in such networks correspond to the semantics of data memories. Analogously, the semantics of transitions from Petri nets and data flows and functions from data-flow diagrams should be considered equivalent.

History

The DFD notation draws on graph theory, originally used in operational research to model workflow in organizations. DFD originated from the Activity Diagram used in the SADT (Structured Analysis and Design Technique) methodology at the end of the 1970s. DFD popularizers include Edward Yourdon, Larry Constantine, Tom DeMarco, Chris Gane and Trish Sarson.

Data-flow diagrams (DFD) quickly became a popular way to visualize the major steps and data involved in software-system processes. DFDs were usually used to show data flow in a computer system, although they could in theory be applied to business process modeling. DFDs were useful to document the major data flows or to explore a new high-level design in terms of data flow.

DFD components



Data flow diagram -

DFD consists of processes, flows, warehouses, and terminators. There are several ways to view these DFD components.[4]

Process

The process (function, transformation) is part of a system that transforms inputs to outputs. The symbol of a process is a circle, an oval, a rectangle or a rectangle with rounded corners (according to the type of notation). The process is named in one word, a short sentence, or a phrase that is clearly to express its essence.

Data Flow

Data flow (flow, dataflow) shows the transfer of information (sometimes also material) from one part of the system to another. The symbol of the flow is the arrow. The flow should have a name that determines what information (or what material) is being moved. Exceptions are flows where it is clear what information is transferred through the entities that are linked to these flows. Material shifts are modeled in systems that are not merely informative. Flow should only transmit one type of information (material). The arrow shows the flow direction (it can also be bi-directional if the information to/from the entity is logically dependent - e.g. question and answer). Flows link processes, warehouses and terminators.

Warehouse

The warehouse (datastore, data store, file, database) is used to store data for later use. The symbol of the store is two horizontal lines, the other way of view is shown in the DFD Notation. The name of the warehouse is a plural noun (e.g. orders) - it derives from the input and output streams of the warehouse. The warehouse does not have to be just a data file, for example, a folder with documents, a filing cabinet, and optical discs. Therefore, viewing the warehouse in DFD is independent of implementation. The flow from the warehouse usually represents the reading of the data stored in the warehouse, and the flow to the warehouse

usually expresses data entry or updating (sometimes also deleting data). Warehouse is represented by two parallel lines between which the memory name is located (it can be modelled as a UML buffer node).

Terminator

The Terminator is an external entity that communicates with the system and stands outside of the system. It can be, for example, various organizations (eg a bank), groups of people (e.g. customers), authorities (e.g. a tax office) or a department (e.g. a human-resources department) of the same organization, which does not belong to the model system. The terminator may be another system with which the modeled system communicates.

Rules for creating DFD

Entity names should be comprehensible without further comments. DFD is a system created by analysts based on interviews with system users. It is determined for system developers, on one hand, project contractor on the other, so the entity names should be adapted for model domain or amateur users or professionals. Entity names should be general (independent, e.g. specific individuals carrying out the activity), but should clearly specify the entity. Processes should be numbered for easier mapping and referral to specific processes. The numbering is random, however, it is necessary to maintain consistency across all DFD levels (see DFD Hierarchy). DFD should be clear, as the maximum number of processes in one DFD is recommended to be from 6 to 9, minimum is 3 processes in one DFD. The exception is the so-called contextual diagram where the only process symbolizes the model system and all terminators with which the system communicates.